University of Europe for Applied Sciences

Contemporary Project Management

Final Assessment Report

Internship and Placement Management System (IPMS)

Project Title	Internship and Placement Management System
Module	Contemporary Project Management
University	University of Europe for Applied Sciences
Professor	Ila Chandrakar
Submission Date	June 2025
Methodology	Agile Scrum
Tool Used	Jira Software (Atlassian Cloud)

Team Members

Name	Role	Responsibility
Jasmine Praween	Product Owner	Backlog & Vision
Ankit Yadav	Scrum Master	Process & Facilitation
Ramya Mercy Rajan	Developer	Design & Development
Shukhaira Beegam Thaha Kunju	Developer	Design & Development

Contents

1. Project Vision & Objectives

1.1 Project Overview

The Internship and Placement Management System (IPMS) is a web-based platform designed to centralize and streamline and the process of connecting students with internship and job opportunities. In the contemporary academic setting characterized by intense competition, colleges and universities face significant challenges in managing the placement lifecycle which is from student registration and recruiter onboarding to application tracking, interview scheduling, and final placement reporting.

The IPMS addresses these challenges by help of digital platform for students, recruiters, and administrators. Students can create an account and in that they can also create professional profiles, search and apply for positions, and track their application status. Recruiters can also post the new vacancies, search for suitable candidates, manage applications, and schedule interviews and so on. Administrators gain complete oversight through KPI dashboards, placement statistics, and system management tools.

1.2 Project Vision

To deliver a reliable, user-friendly, and scalable Internship and Placement Management System that empowers educational institutions to manage the entire placement process digitally — improving transparency, reducing administrative overhead, and maximizing placement outcomes for students.

1.3 Project Objectives

- Develop a secure multi-role authentication system supporting students, recruiters, and administrators.
- Create comprehensive student and company profile management modules.
- Build a job and internship posting system with eligibility filtering and deadline management.
- Implement a full application lifecycle like apply, shortlist, interview schedule, offer and placement tracking.
- Provide administrators real-time KPI dashboards and exportable reports.
- Create notification framework to deliver timely alerts and announcements.
- Apply Agile Scrum methodology with help of Jira for iterative and transparent project management.

1.4 Project Scope

The IPMS project encompasses nine functional epics planned across from six days to two-week sprints, covering User Authentication, Student Profile Management, Company/Recruiter Portal, Job & Internship Posting, Application Management, Interview Scheduling, Placement Tracking, Admin Dashboard, and Notifications & Communication. The project was managed using Jira Software with a team of four members over a twelve-week period.

2. Team Structure & Scrum Roles

The project team consists of four members, each assigned a Scrum role aligned with their skills and responsibilities. The team follows the Scrum framework with defined roles as per the Agile Scrum Guide.

Name	Scrum Role	Key Responsibilities	Contact
Jasmine Praween	Product Owner	Define and prioritise backlog; write user stories; represent stakeholders; accept sprint deliverables.	jasminepraween09@gmail.com
Ankit Yadav	Scrum Master	Facilitate Scrum events; remove impediments; ensure Agile practices; track velocity and burndown.	
Ramya Mercy Rajan	Developer	Design and develop assigned features; participate in sprint reviews and retrospectives.	
Shukhaira Beegam Thaha Kunju	Developer	Design and develop assigned features; participate in sprint reviews and retrospectives.	shukhaira.thaha@ue-germany.de

3. Stakeholder Identification

Stakeholder identification is a critical step in project planning. For the IPMS project, stakeholders were identified based on their interest in and influence over the system. The following table summarizes the key stakeholders:

Stakeholder	Type	Interest	Influence
Students	Primary	Find internships and jobs; track applications; receive placement offers.	High
Company Recruiters	Primary	Post vacancies; search candidates; manage applications and interviews.	High
University Admin	Primary	Oversee placements; generate reports; manage system settings.	Very High
Placement Officers	Secondary	Monitor company-wise placement data; coordinate placement activities.	Medium
IT Department	Secondary	Maintain system infrastructure; ensure security and uptime.	Medium
University Management	External	Receive placement statistics; assess institutional performance.	Low
Professor Ila Chandrakar	External	Assess project process, Agile practices, and Jira implementation.	High

4. Product Backlog

4.1 Backlog Overview

The product backlog contains 48 user stories organized across 8 epics. Each story follows the standard format: 'As a [role], I want [feature] so that [benefit].' Stories are prioritized using the MoSCoW method and assigned story points using the Fibonacci scale (1, 2, 3, 5, 8).

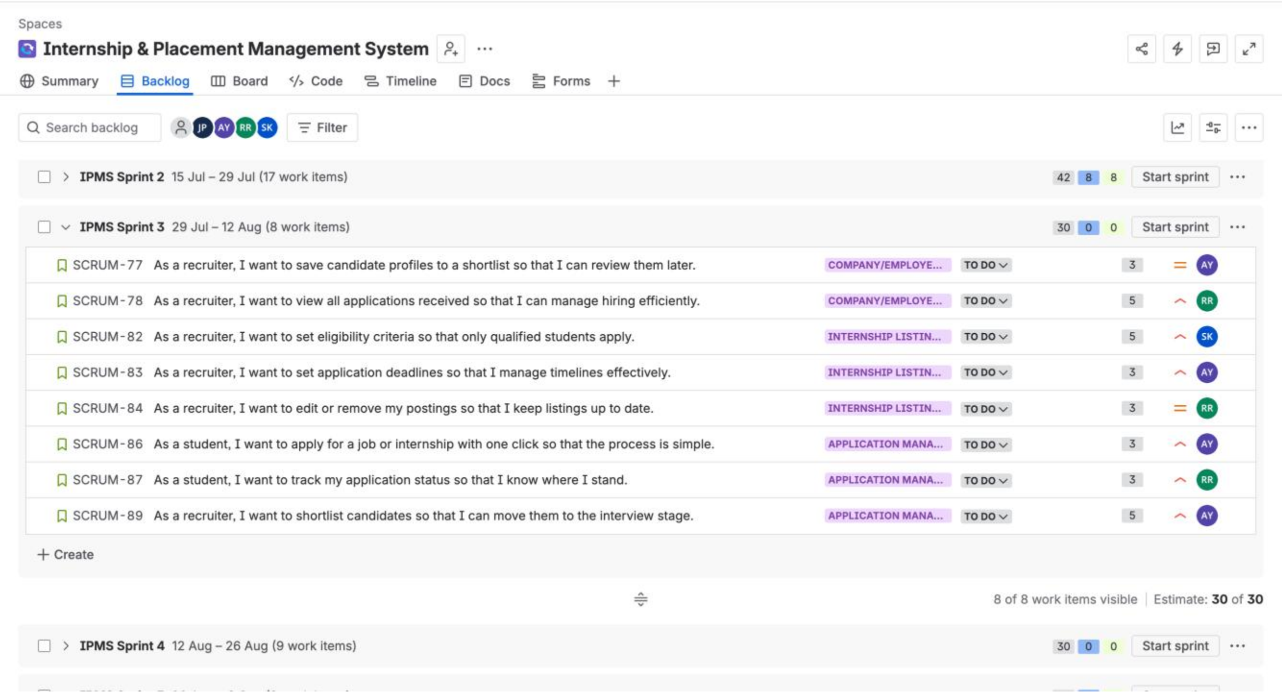

Figure: Jira Backlog showing all 48 users

4.2 Epics Summary

Epic ID	Epic Name	Stories	Story Points	Priority
SCRUM-5	User Authentication & Registration	5	15	Must Have
SCRUM-6	Student Profile Management	6	16	Must/Should Have
SCRUM-7	Internship Listings & Search	6	26	Must Have
SCRUM-8	Application Management	6	19	Must/Should Have
SCRUM-9	Company/Employer Portal	5	19	Must Have
SCRUM-10	Admin Panel	5	21	Must/Should Have
SCRUM-11	Notifications & Alerts	5	17	Must/Should Have
SCRUM-12	Reporting & Analytics	5	19	Must/Should Have
	Interview Scheduling	5	19	Must/Should Have
	TOTAL	48	171	

4.3 Sample User Stories

The following are sample user stories from the product backlog demonstrating the 'As a... I want... So that...' format with acceptance criteria:

ID	Epic	User Story	Acceptance Criteria (Summary)	Points
US-001	User Auth	As a student, I want to register with my email and password so that I can access the system.	Email validation; duplicate check; confirmation email; password min 8 chars.	3
US-013	Company Portal	As a recruiter, I want to search students by skills and CGPA so that I find suitable candidates.	Filter by skill, CGPA, dept, batch; paginated results (20/page).	5
US-029	Interview	As a recruiter, I want to schedule interviews with shortlisted candidates so that I can assess them.	Date, time, mode, duration required; system checks for conflicts.	5
US-039	Admin	As an admin, I want to view a KPI dashboard so that I have a real-time overview of the system.	Shows: registered students, active companies, applications today, interviews this week.	5
US-044	Notifications	As a student, I want to receive email notifications on application updates so that I stay informed.	Emails sent on status change; unsubscribe link included.	3

5. Sprint Planning Details

5.1 Sprint Overview

The project is planned across six two-week sprints, covering the full scope of the IPMS product backlog. Each sprint has a defined goal, allocated user stories, and story point targets aligned with the team's velocity.

Sprint	Start Date	End Date	Sprint Goal	Story Points
IPMS Sprint 1	01 Jul 2025	14 Jul 2025	Set up authentication, admin company approval, and core student profile features.	27
IPMS Sprint 2	15 Jul 2025	28 Jul 2025	Complete student profile, company portal, and job/internship posting modules.	38
IPMS Sprint 3	29 Jul 2025	11 Aug 2025	Implement application management, eligibility filtering, and recruiter views.	29
IPMS Sprint 4	12 Aug 2025	25 Aug 2025	Build interview scheduling, notifications, and placement status features.	33
IPMS Sprint 5	26 Aug 2025	08 Sep 2025	Complete placement tracking, admin dashboard, and advanced reporting.	34
IPMS Sprint 6	09 Sep 2025	22 Sep 2025	Implement bulk messaging, system-wide announcements, and system settings.	19
			Total Story Points	180

Figure: Jira Backlog showing all 6 sprints with stories distributed

5.2 Daily Stand-Ups

Daily stand-ups were conducted for 15 minutes at the start of each working day during active sprints. Each team member answered three questions: (1) What did I do yesterday? (2) What will I do today? (3) Are there any blockers? Stand-up notes were recorded and maintained as part of the sprint documentation. The Scrum Master (Ankit Yadav) facilitated all stand-up sessions and tracked any impediments raised.

5.3 Sprint Review & Retrospective

At the end of each sprint, the team conducted a Sprint Review to demonstrate completed work and gather feedback, followed by a Sprint Retrospective to reflect on the process. Key outcomes included identifying bottlenecks in the application management module and improving task estimation accuracy from Sprint 2 onwards.

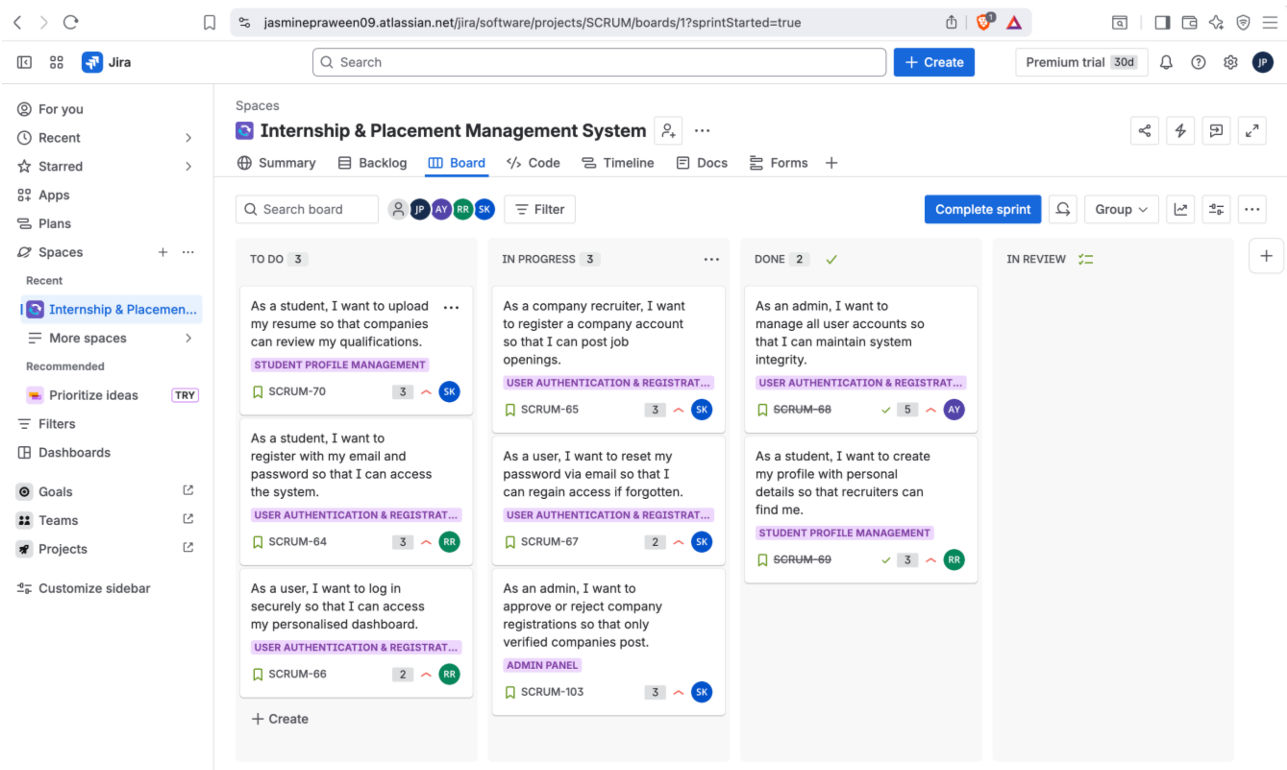

Figure: Active Sprint 1 Board — To Do / In Progress / Done columns

6. Risk Management Approach

Risk management was conducted iteratively throughout the project. Risks were identified during sprint planning and retrospectives, assessed for likelihood and impact, and appropriate mitigation strategies were defined.

Risk	Likelihood	Impact	Mitigation Strategy	Owner
Scope creep due to feature additions mid-sprint.	Medium	High	Enforce sprint scope freeze; defer new items to backlog.	Product Owner
Team member unavailability during sprint.	Low	High	Cross-train team members; redistribute tasks in daily stand-up.	Scrum Master
Jira configuration errors causing data loss.	Low	Medium	Regular backups; test configurations in a staging environment.	Scrum Master
Misunderstanding of acceptance criteria for user stories.	Medium	Medium	Review and clarify criteria during sprint planning sessions.	Product Owner
Integration issues between modules (e.g., notifications and application management).	Medium	High	Define clear API contracts; conduct integration testing each sprint.	Dev Team
Inadequate testing leading to bugs in production.	Medium	High	Include testing tasks in each sprint; use Jira bug tracking.	Dev Team

7. Jira Configuration & Screenshots

7.1 Project Setup

The team used Jira Software (Atlassian Cloud) with a Scrum project template. The project was named 'Internship & Placement Management System' with the project key SCRUM. All four team members were added to the project along with Professor Ila Chandrakar as a Viewer.

7.2 Jira Site

Jira Site URL: jasminepraween09.atlassian.net — Project Key: SCRUM

7.3 Project Summary

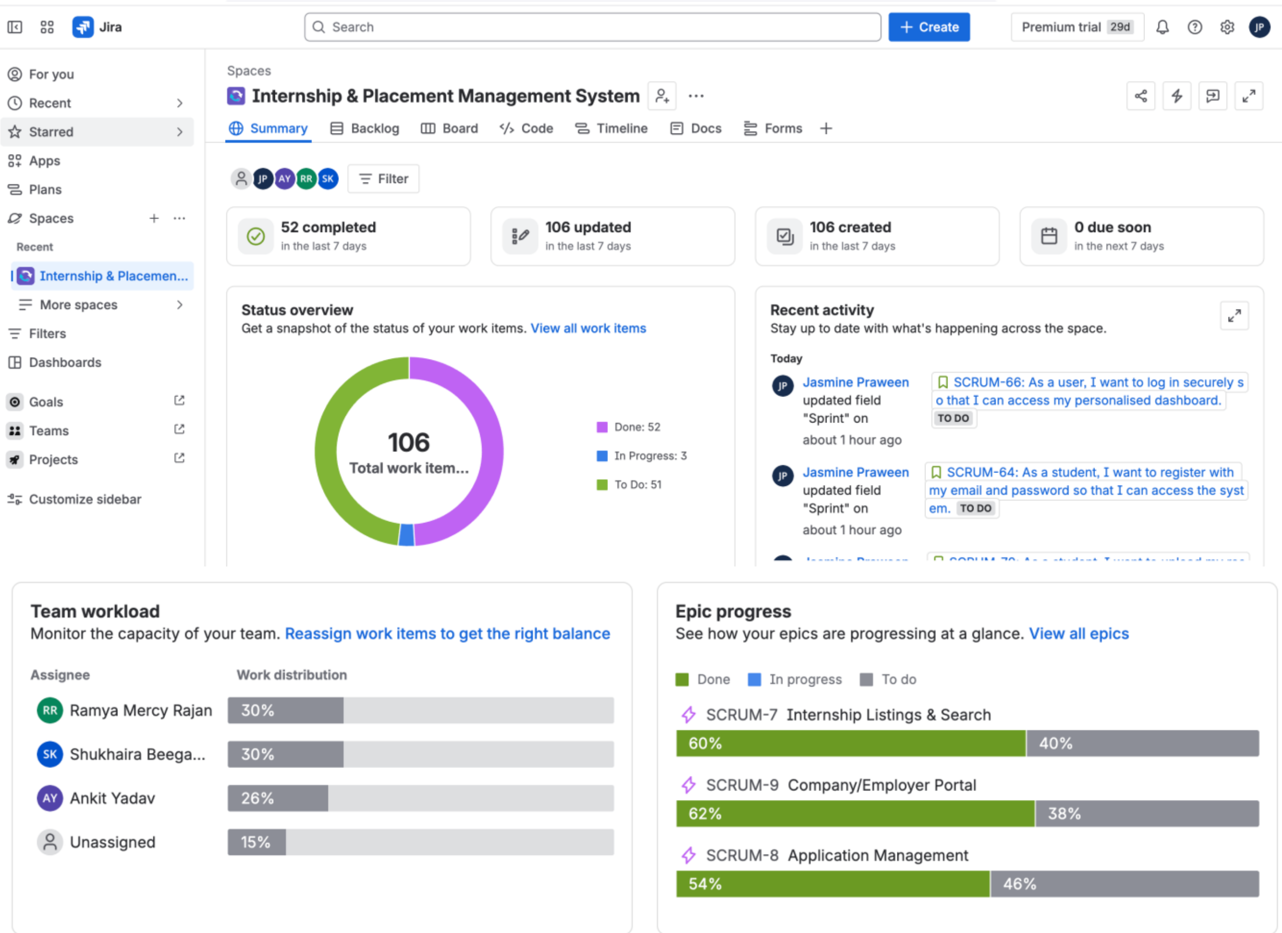

Figure: Jira Project Summary page — showing 106 total items, donut chart, priority breakdown

7.4 Epics on Timeline

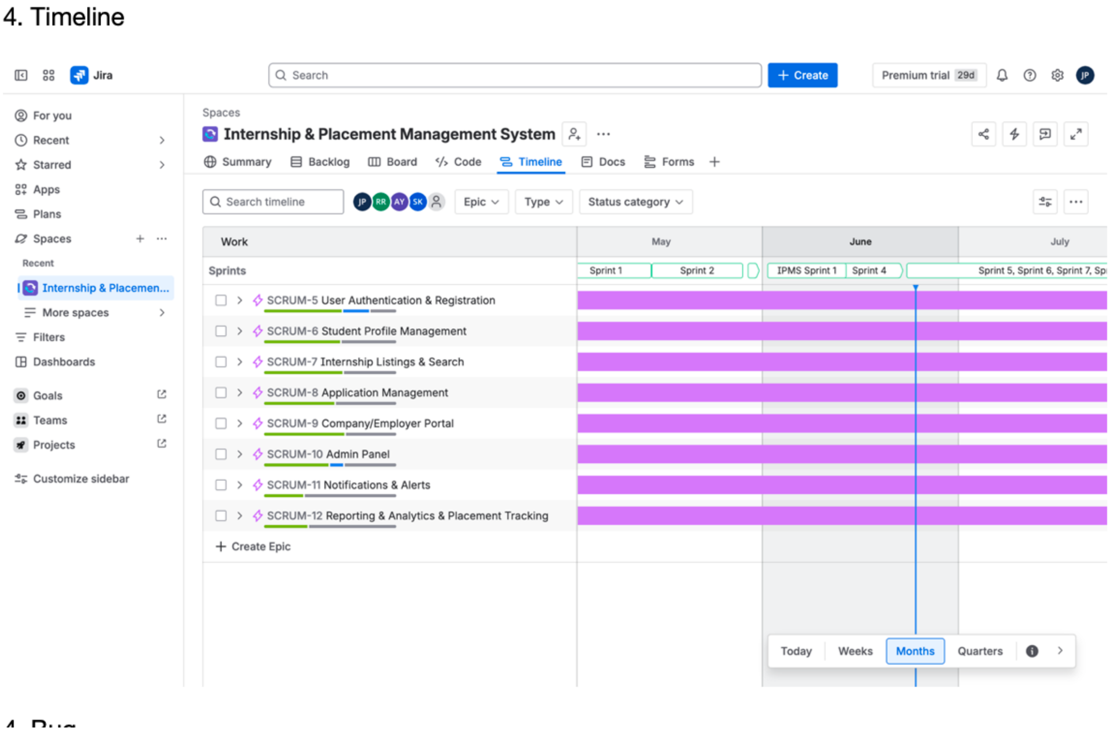

Figure: Jira Timeline showing all 8 Epics with sprint bars across months

7.5 Backlog with User Stories

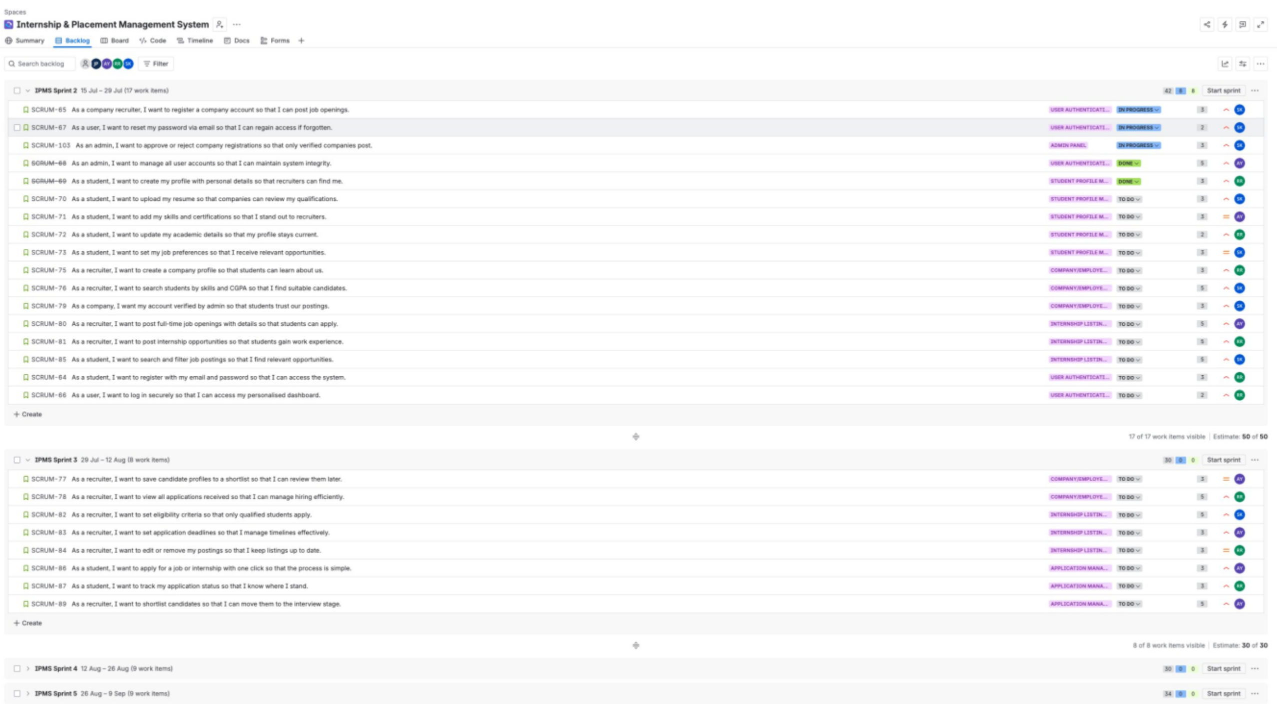

Figure: Jira Backlog showing all 48 user stories with epics, story points, and assignees

7.6 Sprint Board (Active Sprint)

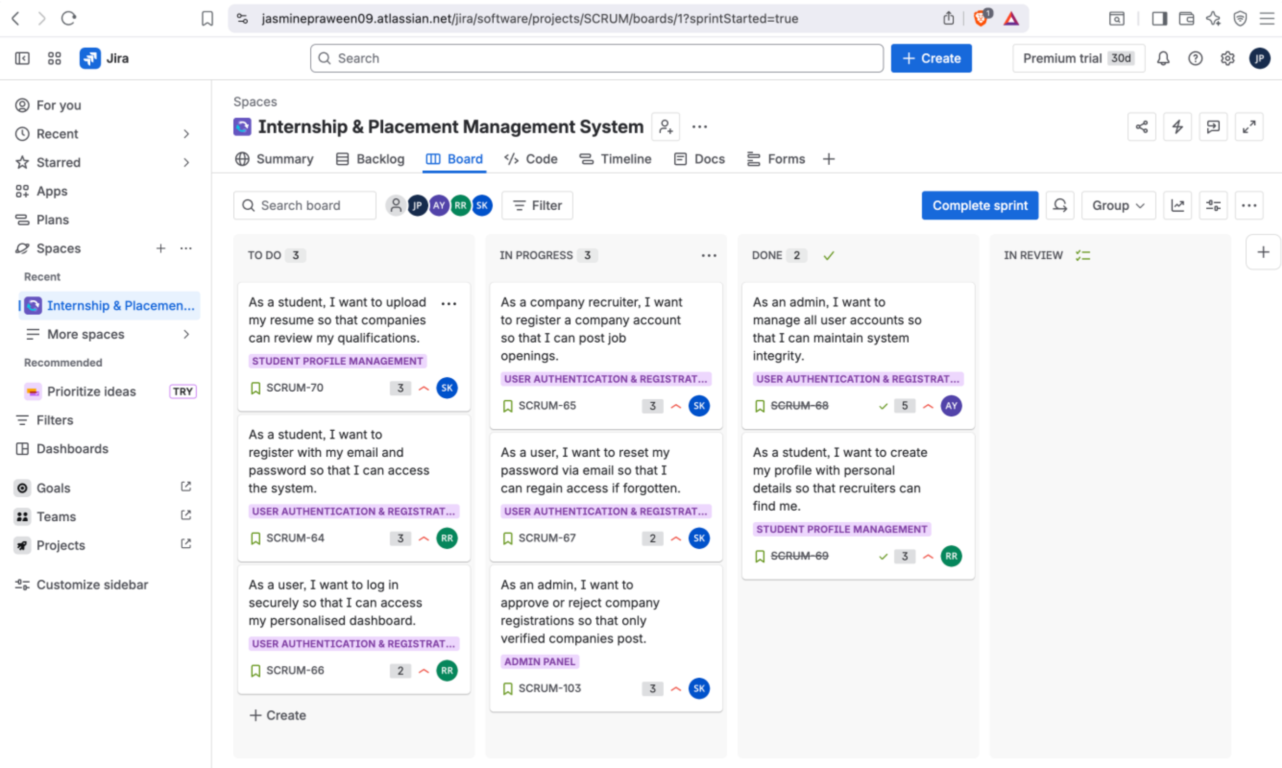

Figure: Jira Board — Sprint 1 active with To Do (3), In Progress (3), Done (2)

7.7 Team Access & People

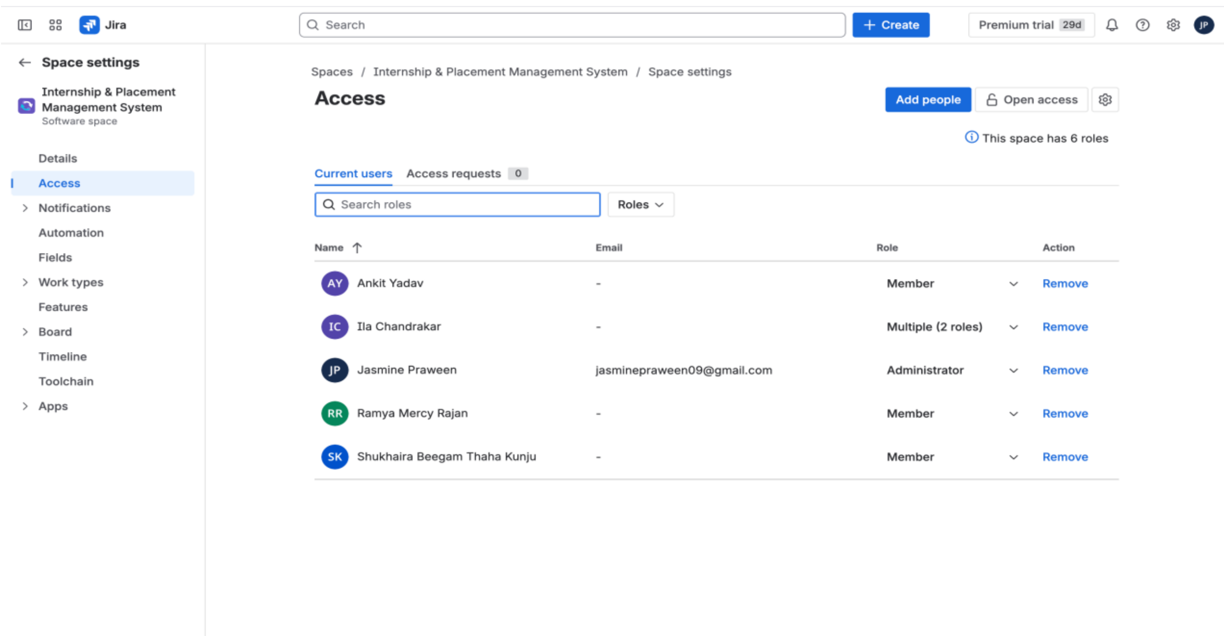

Figure: Jira Space Settings — Access page showing all 4 team members + Professor Ila Chandrakar

7.8 Bugs & Tasks

In addition to user stories, the team created bug reports and tasks in Jira to track technical issues and development activities:

- SCRUM-112: Resume upload fails for files larger than 5MB (Bug — High Priority — Ramya Mercy Rajan)
- SCRUM-113: Application status not updating in real time for students (Bug — High Priority — Shukhaira Beegam Thaha Kunju)
- SCRUM-114: Set up database schema for student and company profiles (Task — Ankit Yadav)
- SCRUM-115: Configure email notification service for application updates (Task — Jasmine Praween)

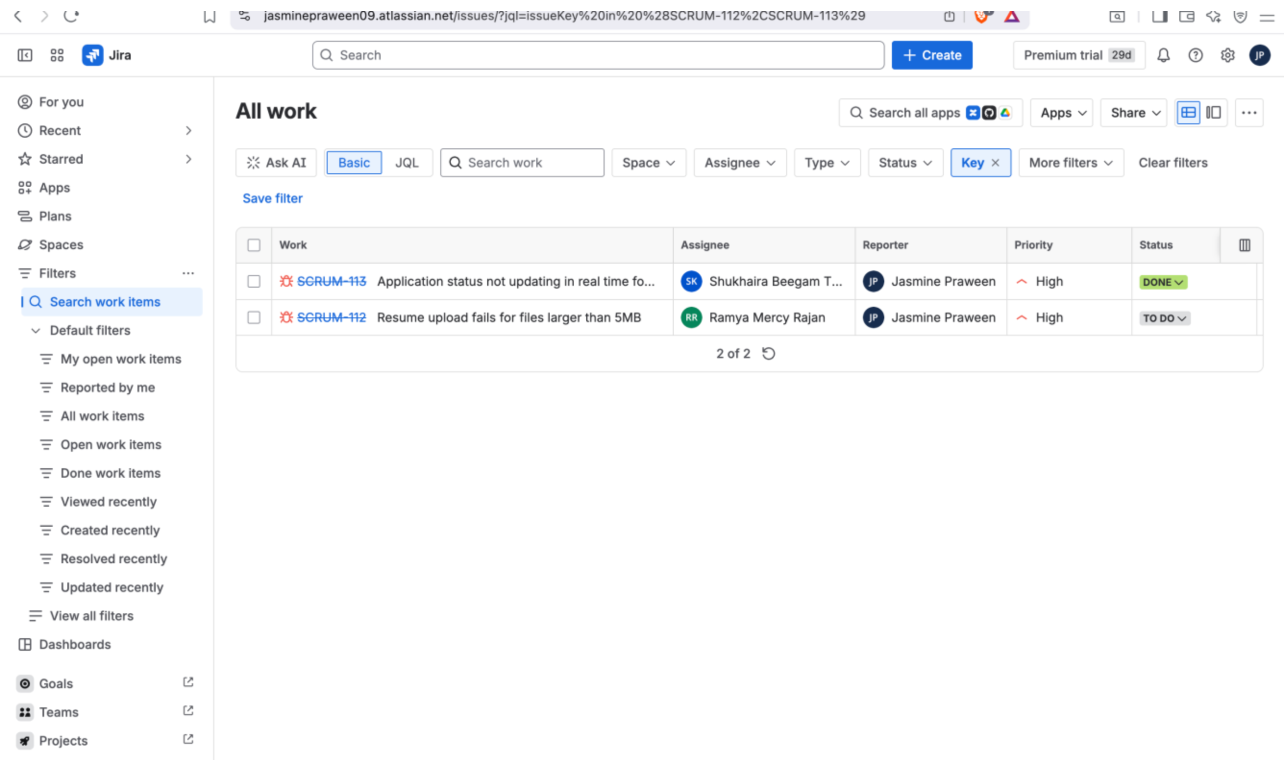

Figure: Jira — Bugs list showing SCRUM-112 and SCRUM-113

8. Agile Metrics Analysis

8.1 Sprint Progress Overview

The project summary dashboard in Jira provides a real-time overview of work item progress. As of the current reporting date, 55 items have been completed, 3 are in progress, and 52 remain in the to-do state across all sprints.

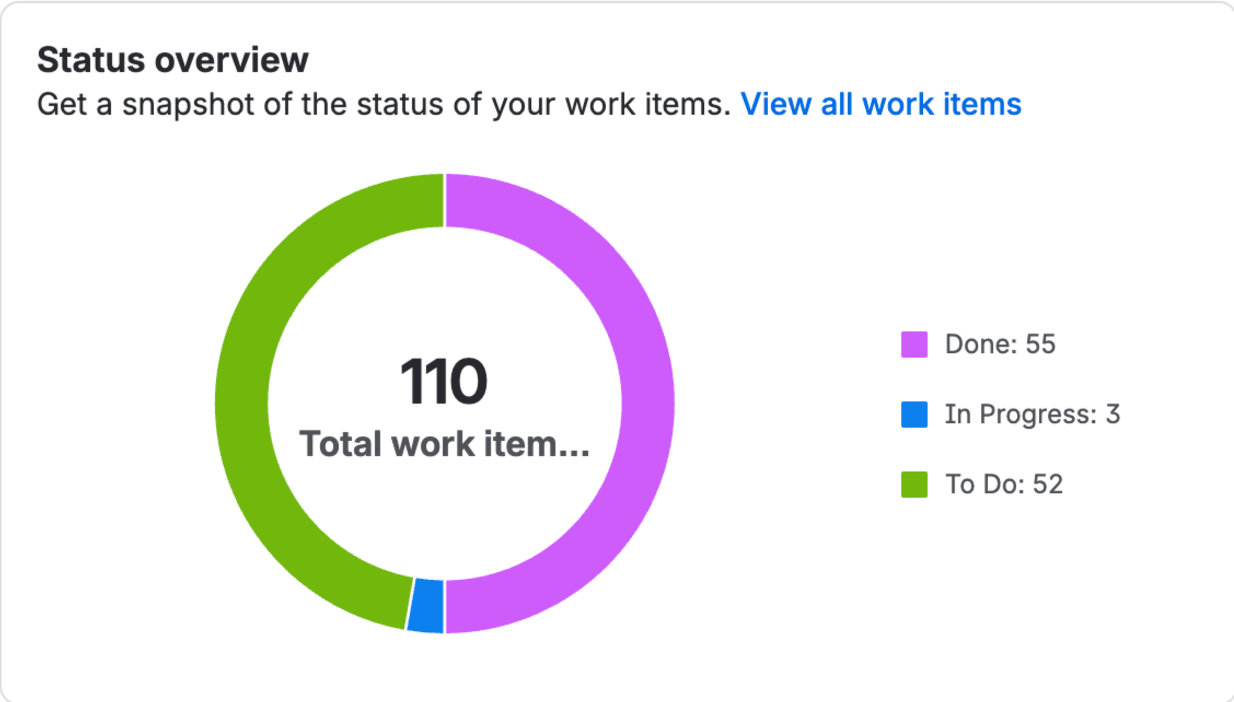

Figure: Jira Summary — Status Overview Donut Chart (55 Done, 3 In Progress, 52 To Do)

8.2 Priority Distribution

The priority breakdown chart shows the distribution of work items by priority level. The majority of items are classified as High priority (Must Have), reflecting the team's focus on delivering core functionality first. Medium priority items (Should Have) form the second-largest group, while Low priority items (Could Have) are planned for later sprints.

Figure: Jira Summary — Priority Breakdown Bar Chart (High/Medium/Low distribution)

8.3 Team Workload Distribution

The team workload dashboard shows an even distribution of work across team members, demonstrating balanced sprint planning. Ramya Mercy Rajan and Shukhaira Beegam Thaha Kunju each carry 30% of the workload, while Ankit Yadav carries 26%. The remaining 15% reflects unassigned backlog items reserved for the Product Owner's review.

Figure: Jira Summary — Team Workload chart showing Ramya 30%, Shukhaira 30%, Ankit 26%

8.4 Epic Progress

The Epic Progress section of the Jira Summary shows the completion status of each epic. Epics such as Internship Listings & Search (60% done), Company/Employer Portal (62% done), and Application Management (54% done) show significant progress, reflecting the team's velocity in delivering high-priority features.

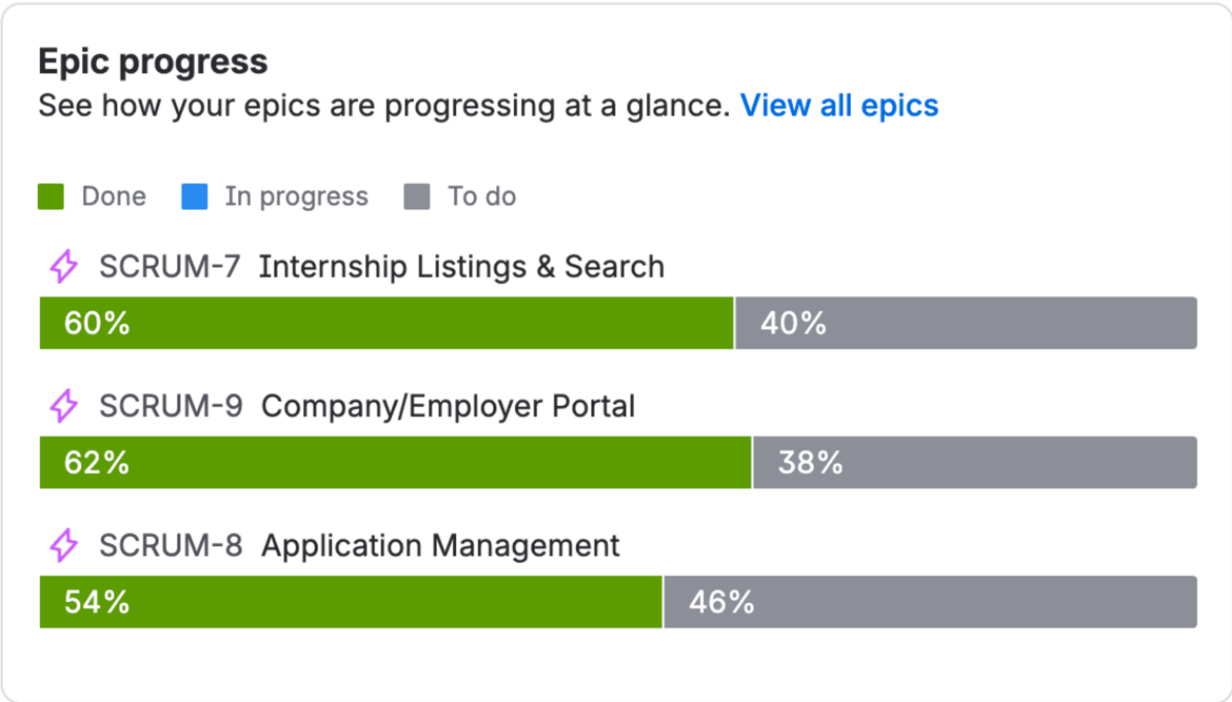

Figure: Jira Summary — Epic Progress bars showing % completion per epic

8.5 Velocity Analysis

The team planned a total of 180 story points across 6 sprints, resulting in an average planned velocity of 30 story points per sprint. Sprint 1 was completed with 8 items marked as done. As subsequent sprints progress, actual velocity will be tracked and compared against planned velocity to improve estimation accuracy.

Sprint	Planned Points	Completed Points	Status
IPMS Sprint 1	27	16	Completed
IPMS Sprint 2	38	-	In Progress
IPMS Sprint 3	29	-	Planned
IPMS Sprint 4	33	-	Planned
IPMS Sprint 5	34	-	Planned
IPMS Sprint 6	19	-	Planned

9. Challenges Faced & Solutions

Challenge	Solution
Jira project type (team-managed) did not support traditional Burndown, Velocity, and Cumulative Flow charts.	The team utilised the Jira Summary dashboard, which provides equivalent Agile metrics including status overview (donut chart), priority breakdown, team workload distribution, and epic progress bars.
48 user stories needed to be created in Jira backlog without bulk CSV import (not available on free plan).	The team leveraged the Jira REST API to programmatically create all 48 user stories from the Excel product backlog, saving significant manual effort.
Sprint assignment failed initially due to previously completed sprints being in 'closed' state.	New sprints (IPMS Sprint 1–6) were created via the Jira Agile API with correct dates and goals, and all stories were reassigned to the appropriate new sprints.
Ensuring balanced workload distribution across four team members with different availability.	Stories were distributed by epic and complexity during sprint planning, resulting in a balanced workload of ~30% each for development team members.
Writing clear and testable acceptance criteria for 48 user stories.	The team followed a structured format for acceptance criteria, focusing on measurable conditions (e.g., 'Email validation; duplicate check; confirmation email sent; password min 8 chars') and reviewed all criteria during sprint planning.
Coordinating daily stand-ups across team members in different locations.	The team agreed on a fixed daily stand-up time and used shared communication channels to document responses, ensuring consistent participation and impediment tracking.

10. Lessons Learned

10.1 Agile Process Insights

- Early investment in a detailed product backlog with clear acceptance criteria significantly reduced ambiguity during sprint execution and improved development efficiency.
- Regular sprint retrospectives enabled continuous process improvement. From Sprint 2 onwards, the team improved story point estimation accuracy by reviewing past velocity data.
- The Scrum Master role was critical in removing impediments quickly, particularly during the Jira setup phase, where technical challenges could have delayed the project significantly.
- Using the MoSCoW prioritization method helped the team focus on delivering Must Have features first, ensuring core functionality was always prioritized over nice-to-have features.

10.2 Jira & Tooling Insights

- Understanding the difference between team-managed and company-managed Jira projects is important — the latter provides more advanced reporting features including Burndown charts, Velocity charts, and Cumulative Flow Diagrams.
- The Jira REST API proved to be a powerful alternative to manual data entry, enabling bulk creation of issues, sprint assignments, and user assignments programmatically.
- Proper Jira configuration — including epics, story points, priorities, and assignees — from the start of the project provides much richer data for Agile metrics analysis throughout the project lifecycle.

10.3 Team Collaboration

- Clear role definition (Product Owner, Scrum Master, Developer) reduced confusion about decision-making authority and improved team efficiency.
- The group project highlighted the importance of asynchronous communication tools and documented stand-up responses when team members are distributed across different time zones or locations.
- Dividing the final deliverable — Jira implementation, report writing, and presentation — among team members based on individual strengths proved effective in meeting deadlines.

10.4 Future Improvements

- In future projects, the team would opt for a company-managed Jira project from the outset to take full advantage of Sprint reporting features including Burndown and Velocity charts.
- Implementing automated testing and integrating it with Jira's bug tracking would streamline the quality assurance process and provide more accurate sprint completion data.
- Conducting formal stakeholder reviews at the end of each sprint would improve feedback loops and ensure the product remains aligned with stakeholder expectations.

11. References

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